NAAN MUDHALVAN PROJECT

TITLE NAME: Courier Management System (CMS)

PROJECT REPORT

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Courier Management System (CMS)

User Story:

Courier and logistics organizations frequently face challenges in managing parcel deliveries efficiently because many operational activities rely on manual records and disconnected systems. Without a centralized tracking mechanism, monitoring the real-time movement of parcels becomes difficult. This results in delayed updates, reduced transparency for customers, and operational inefficiencies.

Delivery agent assignments are often performed manually, which may cause scheduling conflicts and slow down the delivery process. Additionally, the lack of automated communication systems prevents customers from receiving timely shipment notifications. Managers also struggle to evaluate operational performance due to limited analytical insights.

When multiple branches are involved, coordination becomes more complicated without an integrated system. These issues collectively lead to delays, reduced productivity, customer dissatisfaction, and potential financial loss for courier companies.

1. INTRODUCTION

1.1 Project Overview:

The Courier Management System (CMS) is a system designed to manage courier and parcel delivery operations in an organized way. It helps in recording courier details, managing sender and receiver information, and tracking the delivery status of parcels.

This system reduces manual work by storing all courier data digitally and making it easy to update and access. It also allows administrators to monitor courier activities and maintain accurate records.

The project is developed using the Salesforce platform, which helps in creating objects, fields, and automation to manage the courier process efficiently. Overall, the system improves efficiency, saves time, and makes courier tracking easier.

1.2 Purpose:

The purpose of the Courier Management System (CMS) is to simplify and automate the process of managing courier services. It helps in storing courier details, tracking parcels, and maintaining sender and receiver information in a digital format.

This system reduces manual work, improves accuracy, and allows users to easily monitor the delivery status of parcels. The project is developed using the **Salesforce platform to manage courier operations more efficiently.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnected	stressed
PS-2	Customer	track parcels	updates are delayed	no centralized system	uncertain

2.2 Empathy Map Canvas

User Persona: Branch Manager

- **Think & Feel:** Concerned about delays & errors
- **Hear:** Customer complaints
- **See:** Registers & spreadsheets
- **Say & Do:** Manual follow-ups
- **Pain:** Delivery conflicts & missed updates
- **Gain:** Automated tracking & dashboards

2.3 Brainstorming & Idea Prioritization

Ideas Considered

- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

Final Selection

Salesforce CRM selected due to:

- Automation
- Scalability
- Real-time visibility

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation
FR-6	Reports & dashboards

Non-Functional Requirements

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram

Flow: Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.

4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce CMS
Innovation	Automated tracking & billing
Impact	Faster deliveries
Business Model	Courier services
Scalability	Multi-branch ready

4.3 Solution Architecture Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

5. PROJECT PLANNING & SCHEDULING

Project Planning

- Identify the requirements of the Courier Management System (CMS).
- Define the main modules such as courier details, sender information, receiver information, and tracking.
- Design the system structure and workflow.
- Create custom objects and fields using the **Salesforce platform.
- Implement automation features like validation rules, workflows, and reports.
- Test the system to ensure all functions work properly.
- Deploy the system and maintain it for smooth operation.

Project Development Phase:

- Create a developer account in Salesforce.
- Set up the application environment for the Courier Management System (CMS).
- Create custom objects such as Courier, Sender, and Receiver.
- Add required fields to store courier and delivery information.
- Configure relationships between the objects.
- Implement validation rules and automation features.
- Test the system to ensure all functions work correctly.
- Deploy the completed project for use.

What you'll learn

- Real-time Salesforce project implementation
- Designing custom objects and relationships
- Creating page layouts & record types
- Writing validation rules
- Building automation with flows
- Developing reports & dashboards
- Setting up security with profiles, permission sets, and sharing
- End-to-end application testing

6. Salesforce Account

Introduction:

A Salesforce account is used to access the Salesforce platform and develop applications. It allows users to create and manage objects, fields, automation, and other features required for building projects.

For this project, a Salesforce account is used to develop the Courier Management System (CMS). Using the platform, we can create custom objects for courier details, manage data, and automate processes such as tracking and updating delivery status. This helps in building and managing the system easily on the cloud.

What Is Salesforce ?

Salesforce is a cloud-based software platform used by organizations to manage customer data, business processes, and applications. It helps companies store information, track activities, and automate tasks in one place.

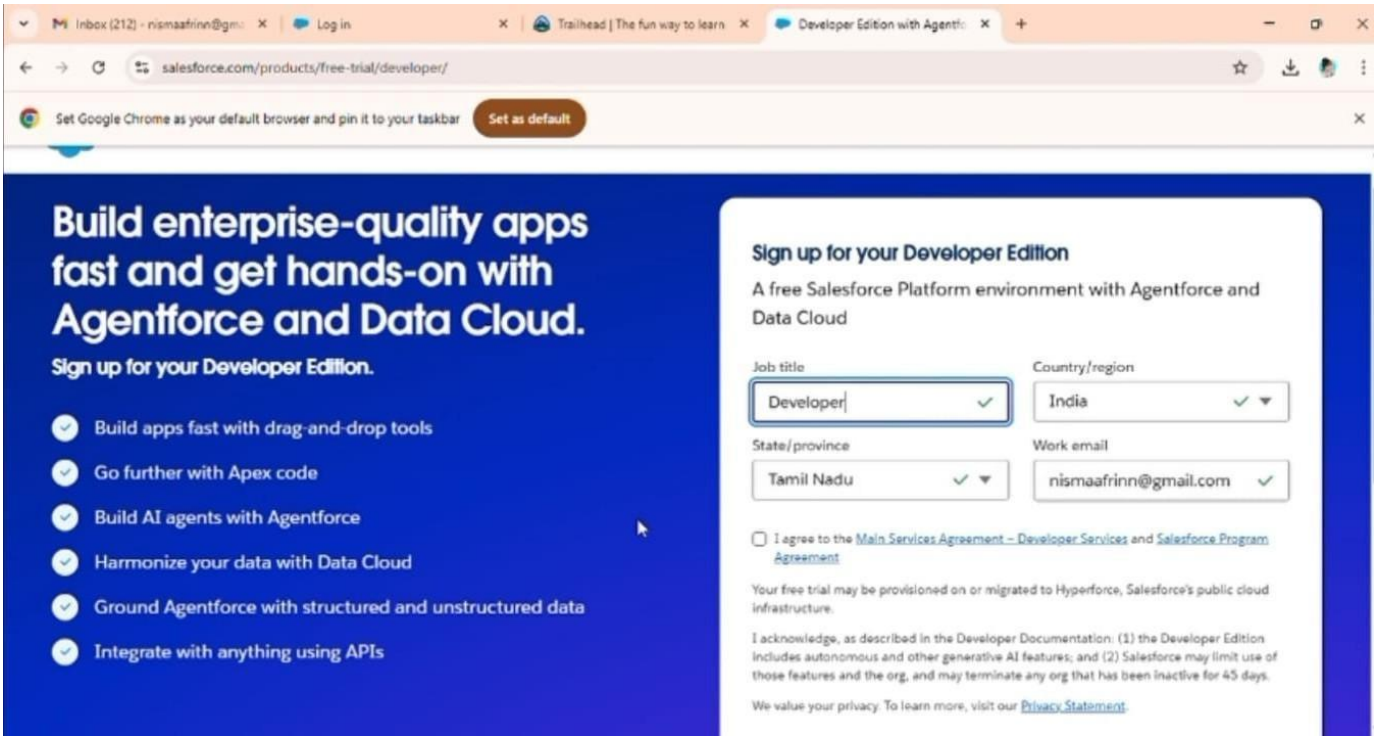
Salesforce allows users to create custom applications, manage data, generate reports, and automate workflows without needing complex programming. Because it works on the cloud, users can access it from anywhere using the internet.

It is widely used for customer relationship management (CRM) and for building systems like the Courier Management System (CMS) to manage and track business operations efficiently

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the signup form, enter the following details :
3. First name & Last name
4. Email

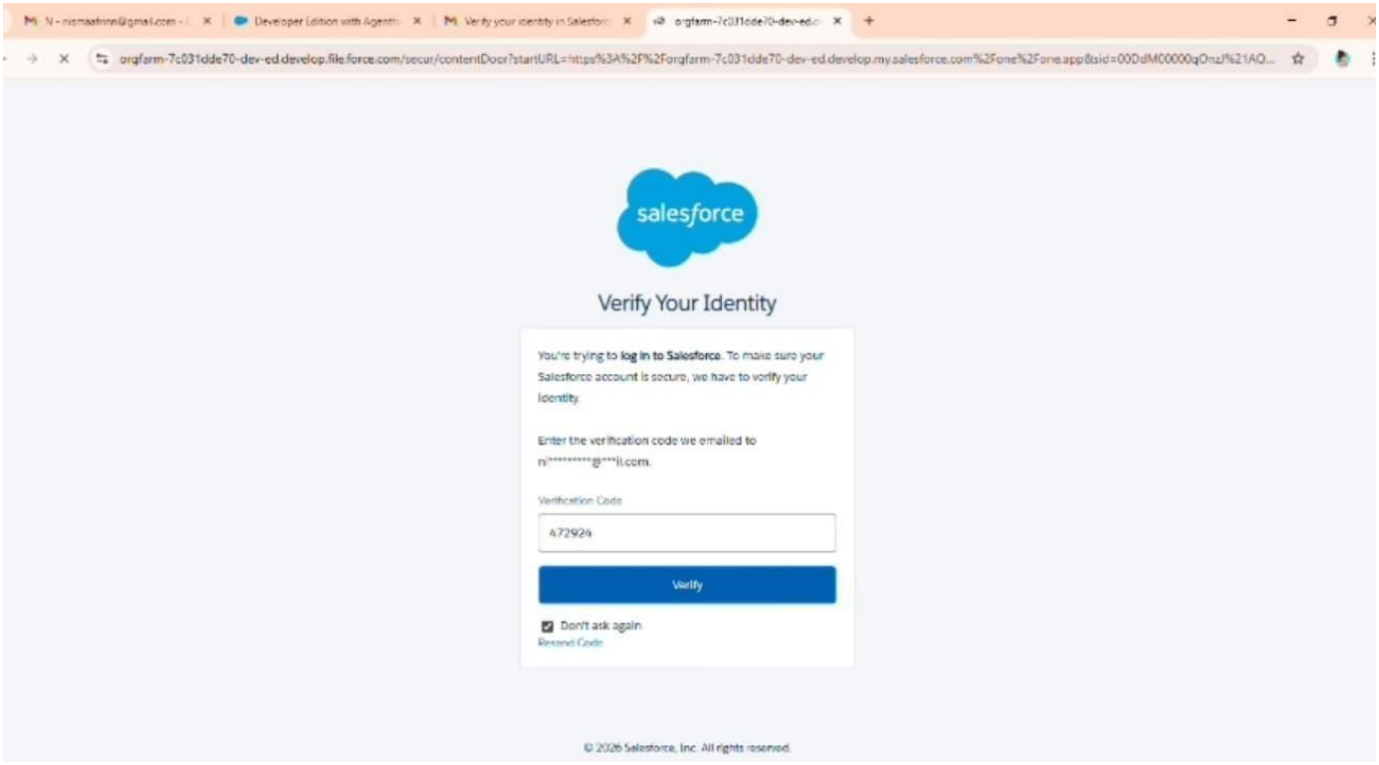


1) First name & Last name
2) Email
3) Job Title: Developer
4) Company: College Name
5) Country: India

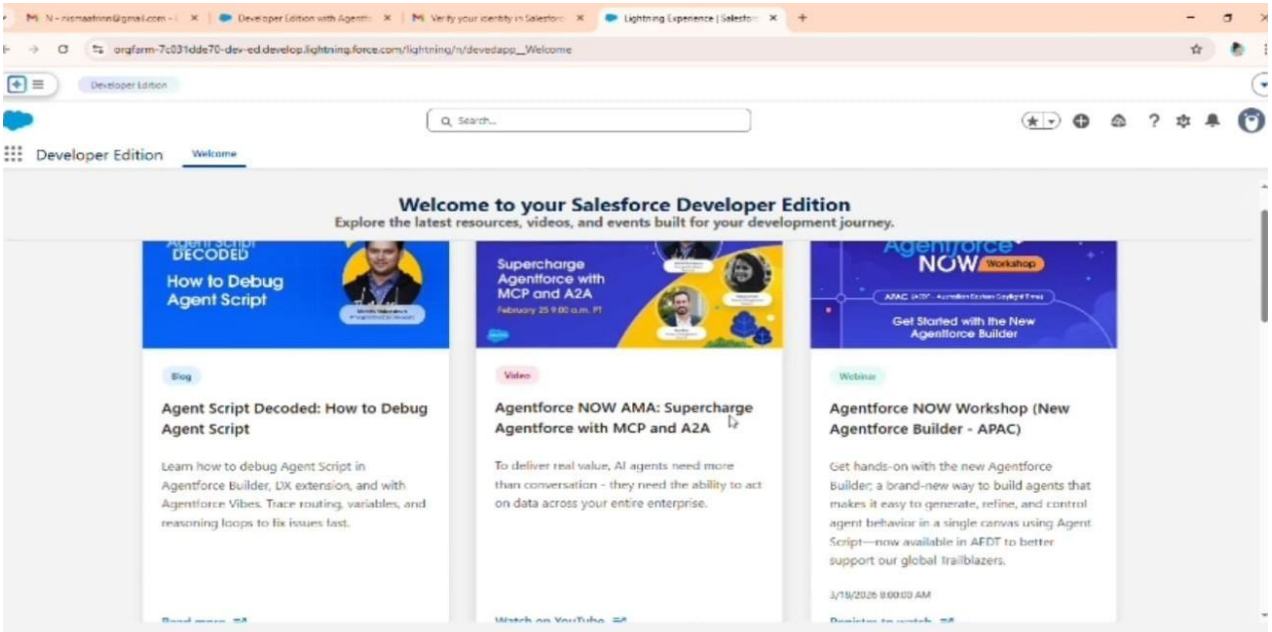
Click on sign me up after filling these.

Activity 2: Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2 hours.



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.
4. Then you will redirect to your salesforce setup page.



7. Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

Custom Objects Created:

1. **Customer__c** – stores customer personal and contact information
2. **Shipment__c** – manages parcel details, source, destination, and tracking information
3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details
4. **Branch__c** – represents courier branch offices
5. **Delivery_Update__c** – stores status updates for each shipment
6. **Invoice__c** – manages billing details for delivered shipments

Relationships:

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

Delivery_Agent → Branch (Lookup)

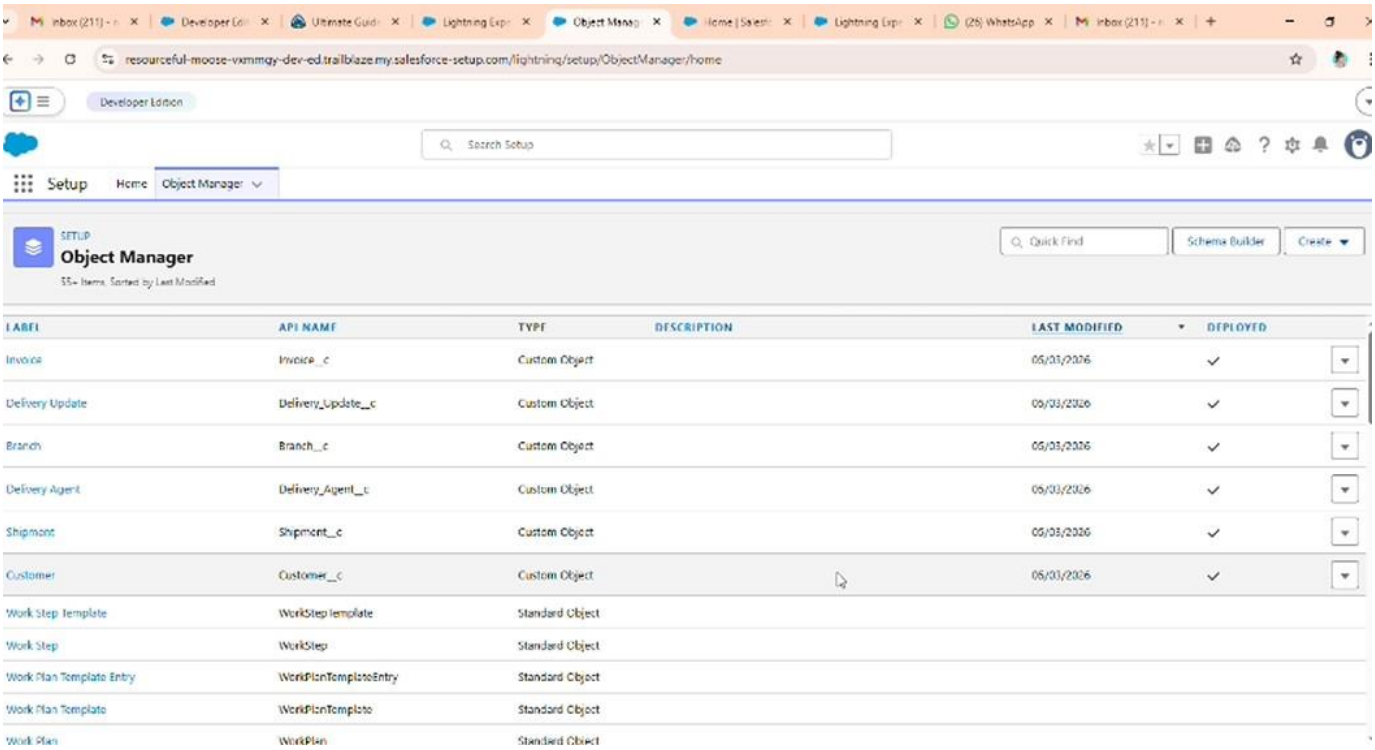
Delivery_Update → Shipment (Master-Detail)

Invoice → Shipment (Lookup)

Steps to Implement :

Activity 1: Create Customer Object

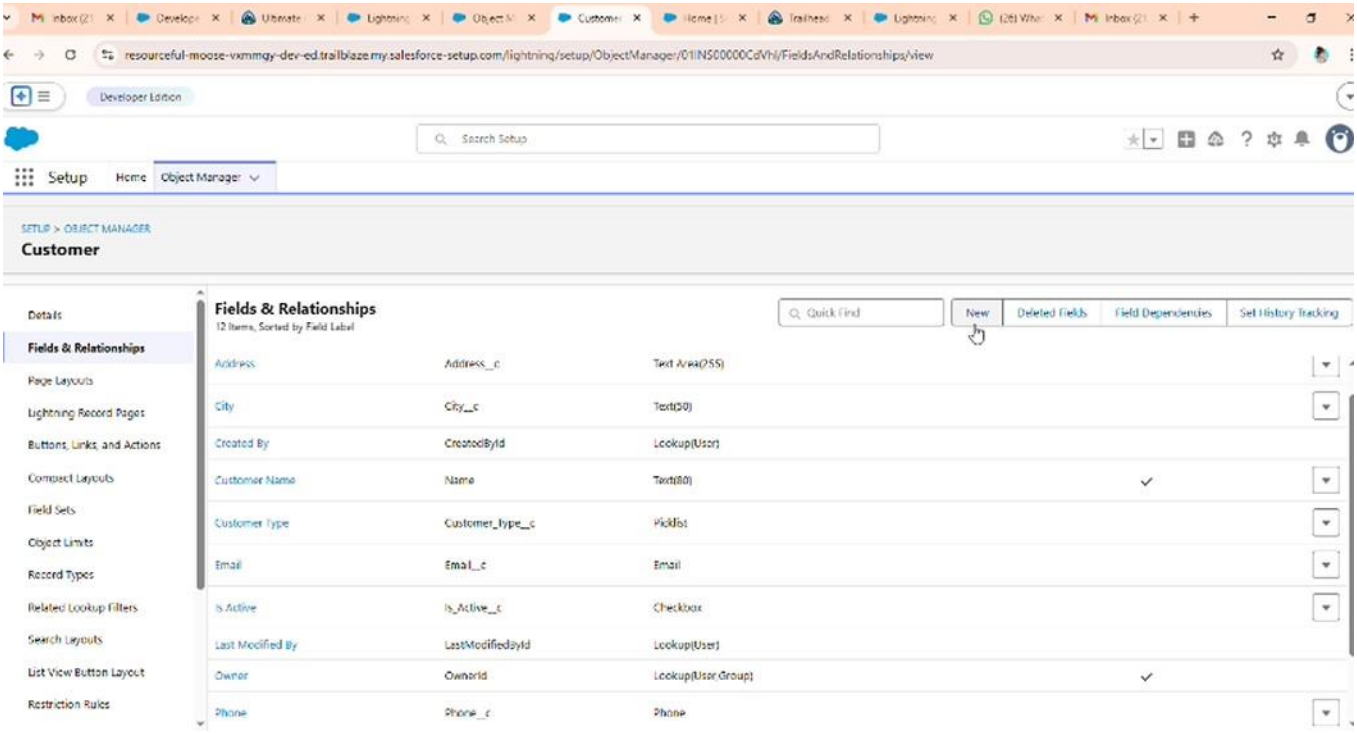
1. Click Setup → Object Manager → Create → Custom Object



2. Enter the following details:

   - Label: Customer
   - Plural Label: Customers
   - Record Name: Customer Name (Text)

3. Under Optional Features:
   - Enable Allow Reports
   - Enable Track Field History
   - In Search Status check the box Allow Search
   - In Object Creation Options check the box Add Notes and Attachments related list to default page layout
   - Click Save.



Activity 2: Create Shipment Object

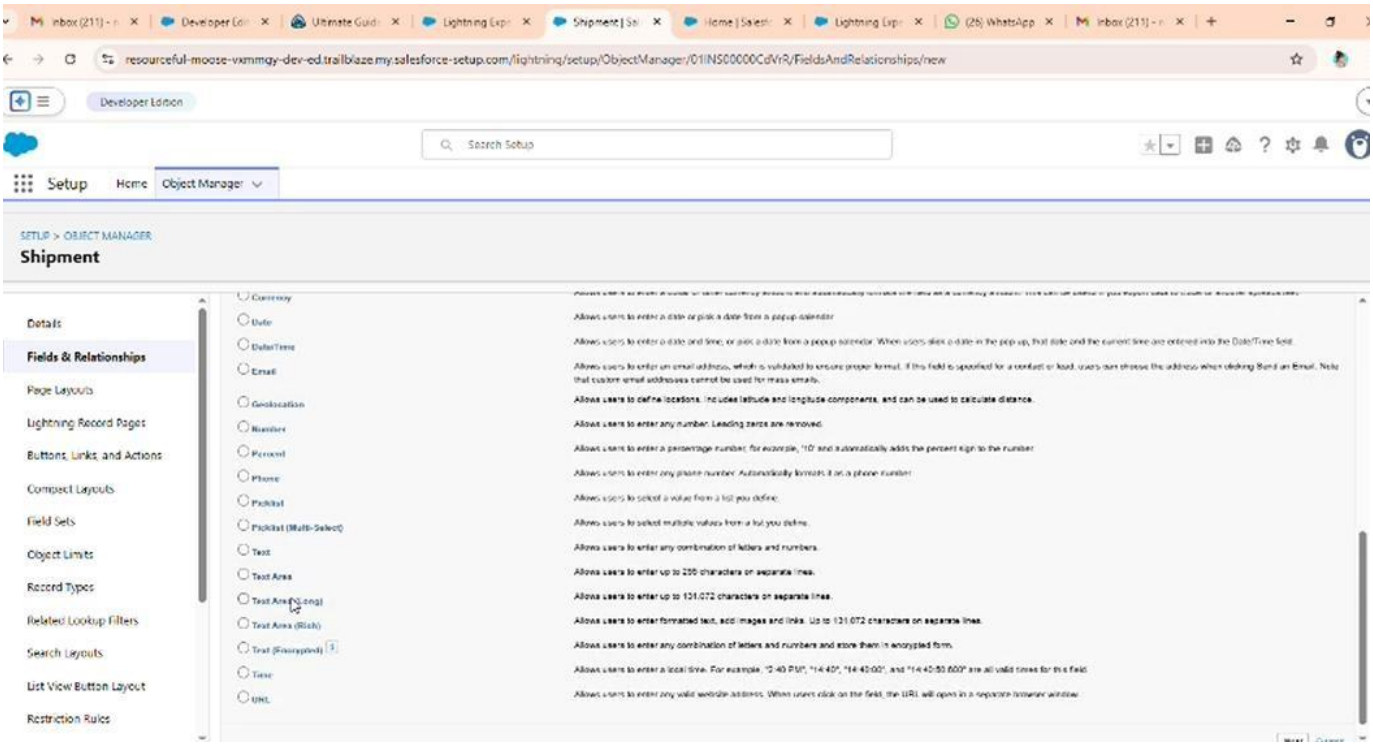
1. Go to Setup → Object Manager → Create → Custom Object

2. Enter:
   - Label: Shipment
   - Plural Label: Shipments
   - Record Name: Tracking Number (Auto Number)

3. Example Format: SHP- {00000}

   Starting Number: 0

4. Enable:
   - Allow Reports
   - Track Field History
5. Click Save



Activity 3: Create Delivery_Agent Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:
    - Label: Delivery Agent
    - Plural Label: Delivery Agents
    - Record Name: Agent Name (Text)

3. Enable:
    - Allow Reports
    - Track Field History

4. Click Save

Activity 4: Create Branch Object

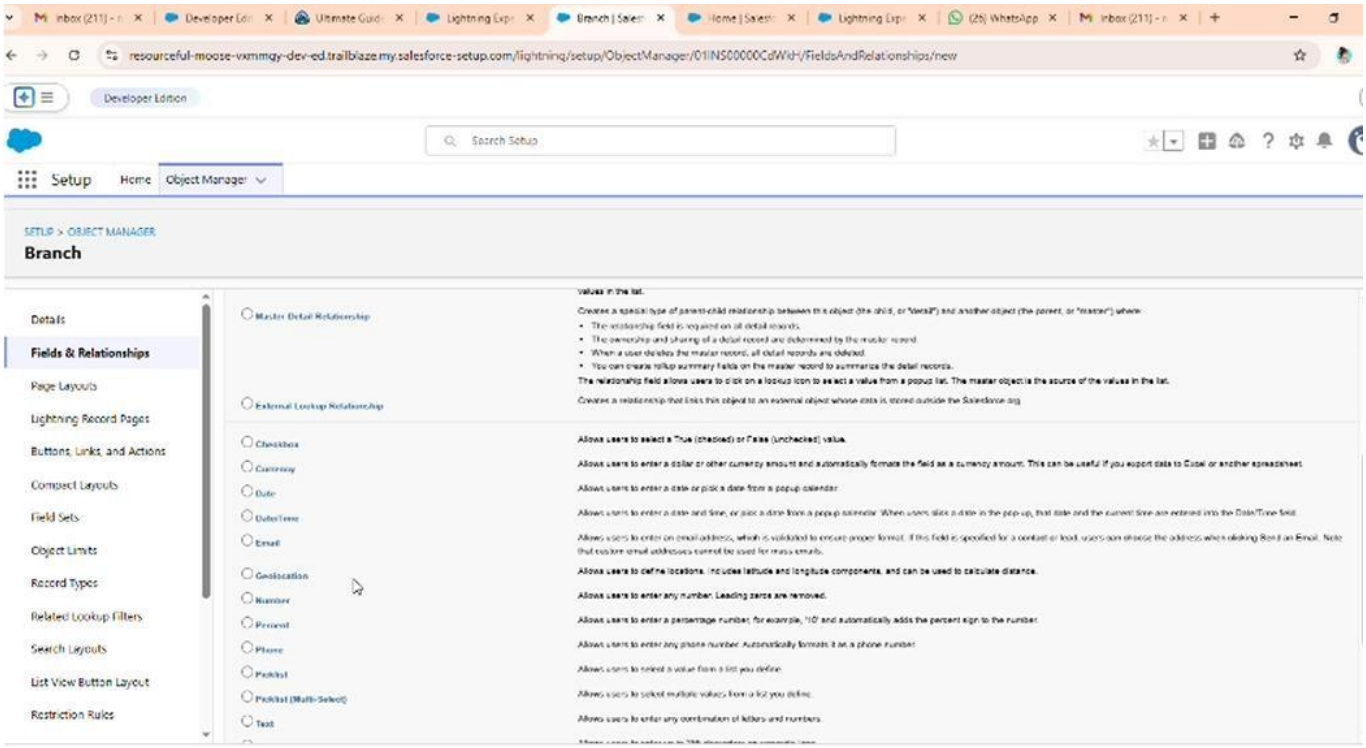
1. Setup → Object Manager → Create → Custom Object

2. Enter:
   - Label: Branch
   - Plural Label: Branches
   - Record Name: Branch Name (Text)

3. Enable:

   Allow Reports

4. Click Save.



Activity 5: Create Delivery_Update Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:
   - Label: Delivery Update
   - Plural Label: Delivery Updates
   - Record Name: Update Number (Auto Number)

3. Example Format: UPD-{0000}

   Enter Starting number: 0

4. Enable:
   - Allow Reports
   - Track Field History

5. Click Save.

Activity 6: Create Invoice Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:
   - Label: Invoice
   - Plural Label: Invoices
   - Record Name: Invoice Number (Auto Number)

3. Example Format: INV- {0000}

   Enter Starting number: 0

4. Enable:
   - Allow Reports
   - Track Field History

5. Click Save

Fields & Relationships

What are fields ?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1: Creation of fields for the Customer Object

Field Name	API Name	Data Type	Required	Description
Customer Name	Name	Text	Yes	Full name of customer
Email	Email__c	Email	Yes	Customer email
Phone	Phone__c	Phone	Yes	Customer contact number
Address	Address__c	Text Area	Yes	Customer address
Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer city
State	State__c	Text	No	Customer state
Pincode	Pincode__c	Text	No	Postal code
Is Active	Is_Active__c	Checkbox	No	Active status

1. Email (Email Field)

- Go to Setup → Object Manager → Customer__c
- Click Fields & Relationships → New
- Select Email → Next
- Field Label: Email
- Check the Required box
- Click Next → Next → Save

2. Phone (Phone Field)

- Setup → Object Manager → Customer__c
- Fields & Relation
- Select Phone → Next
- Field Label: Phone
- Check Required

- Click Next → Next → Save

3. Address (Text Area)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Textarea → Next
- Field Label: Address
- Check Required
- Save

4. Customer Type (Picklist)

- Setup → Customer__c → Fields → New
  - Picklist → Next
  - Field Label: Customer Type
  - Enter values:
    - Regular
    - Business
  - Check Required
  - Save

5. City (Text)

- Setup → Customer__c → Fields → New
  - Text → Next
  - Field Label: **City**
  - Save

6. State (Text)

- Setup → Customer__c → Fields → New
  - Text → Next
  - Field Label: **State**
  - Save

7. Pin code (Text)

- Setup → Customer__c → Fields → New
  - Text → Next
  - Field Label: **Pin code**
  - Save

8. Is Active (Checkbox)

- Setup → Customer__c → Fields → New
  - Checkbox → Next
  - Field Label: **Is Active**
  - Save

8. Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Activity 1: Auto-Update Shipment Status When Delivery Update is Created

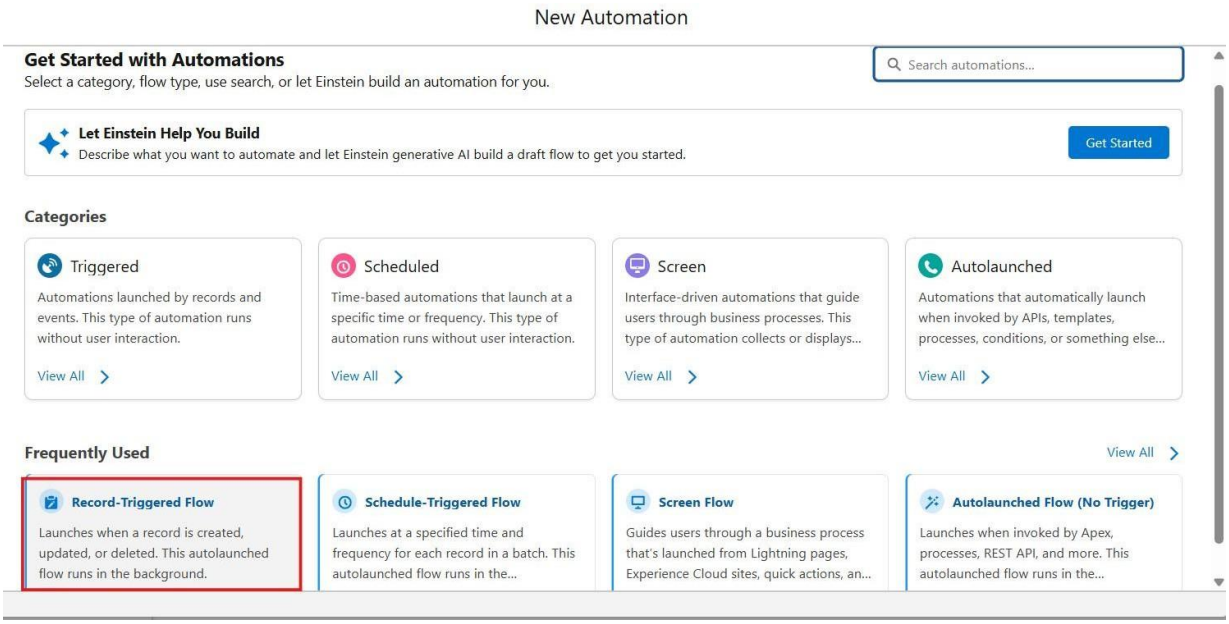
Scenario:

Whenever a Delivery_Update__c record is created, automatically update the related Shipment__c Status with the latest status from Delivery Update.

Part 1: Create the Record-Triggered Flow

Step 1: Open Flow Builder

- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Flows
- Click Flows
- Click the New Flow button
- Select Record-Triggered Flow
- Click Create



Step 2: Configure the Start (Trigger)

On the Start panel on the right:

- Object: select Delivery_Update__c
- Trigger the Flow When:
  - Select A record is created
- Condition Requirements:
  - Choose: None (Always Run the Flow)
- Optimize the Flow For:
  - Select Actions and Related Records
- Click Done

Part 2: Update Related Shipment Record

Step 3: Add "Update Records" Element

- Click the + icon under the Start element
- Select Update Records

Configure the Update Records Element:

- Label: Update_Shipment_Status
- How to Find Records to Update:
- Select Specify conditions to identify records

Set Filter Conditions:

- Object: Shipment__c
- Field: Id
- Operator: Equals
- Value: {!$Record.Shipment__c}

(This means update the Shipment linked to the Delivery Update)

Set Field Values to Update Add field mapping:

Shipment Field	Value Source
Status__c	{!$Record.Status__c}

(This copies status from Delivery Update to Shipment)

- Click Done

Part 3: Save and Activate Step

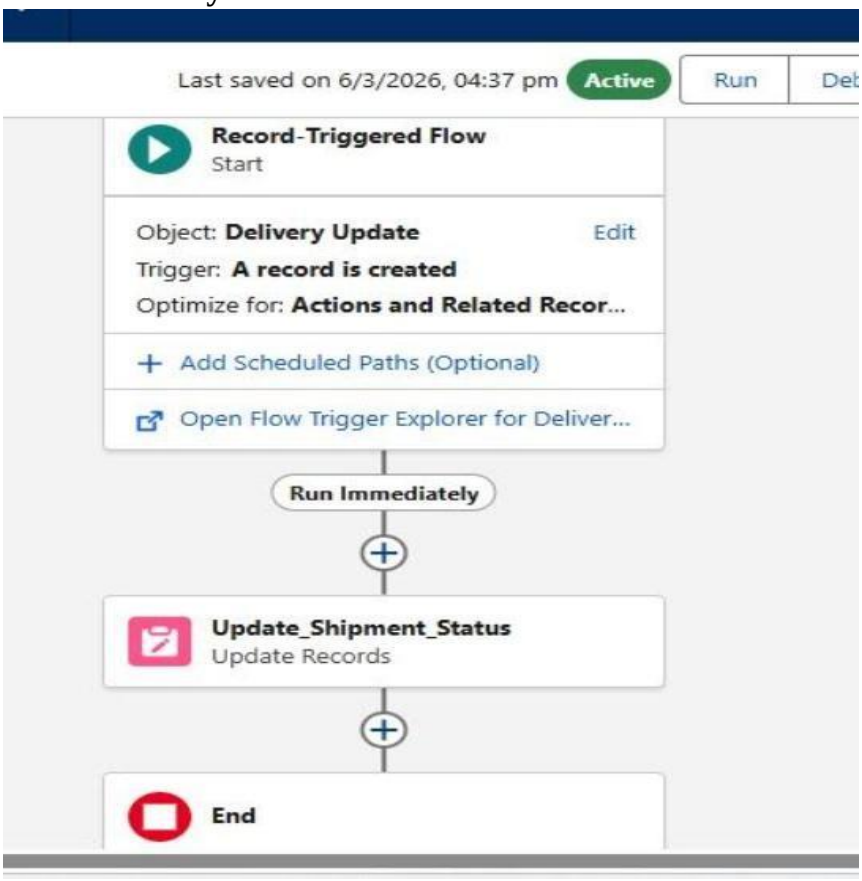
Step 4: Save Flow

- Click Save (top-right)
- Flow Label: Auto_Update_Shipment_Status
- API Name will auto-fill
- Click Save

Step 5: Activate Flow

- Click Activate
- Flow status should show Active

Final Flow Layout:



9. The Lightning App :

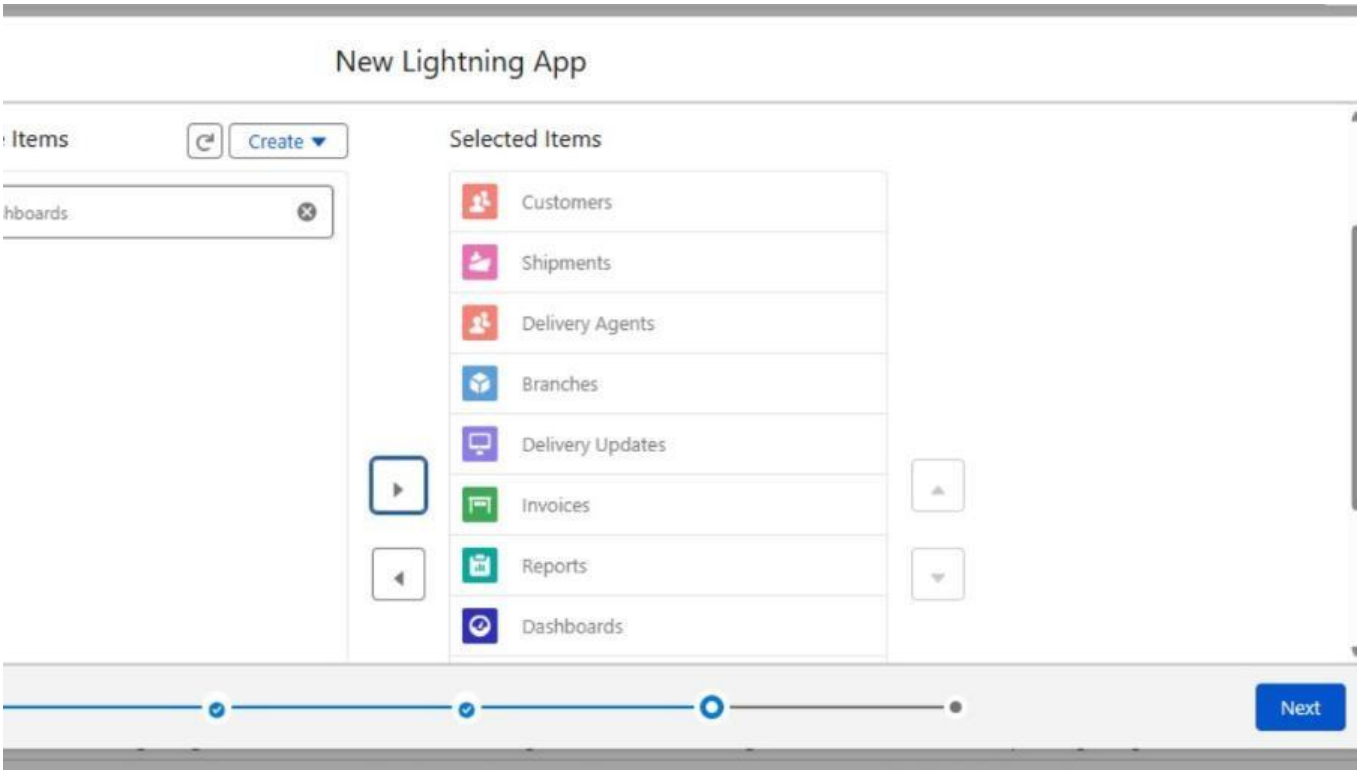
What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1: Create a Lightning App

Create the Courier Management System App

1. From Setup, enter App Manager in the Quick Find and select App Manager.
2. Click New Lightning App.
    - Enter the Courier Management System as the App Name.
3. Upload an image or icon for the app, then click Next.
4. Under App Options, leave the default selections and click Next.
5. Under Utility Items, leave as it is and click Next.



6. From Available Items, select the following and move them to Selected Items:

   - Customer
   - Shipment
   - Delivery Agent
   - Branch
   - Delivery Update
   - Invoice
   - Reports
   - Dashboards

7. Click Next.

8. From Available Profiles, select System Administrator and move it to Selected Profiles.

9. Click Save & Finish.

10. Role:

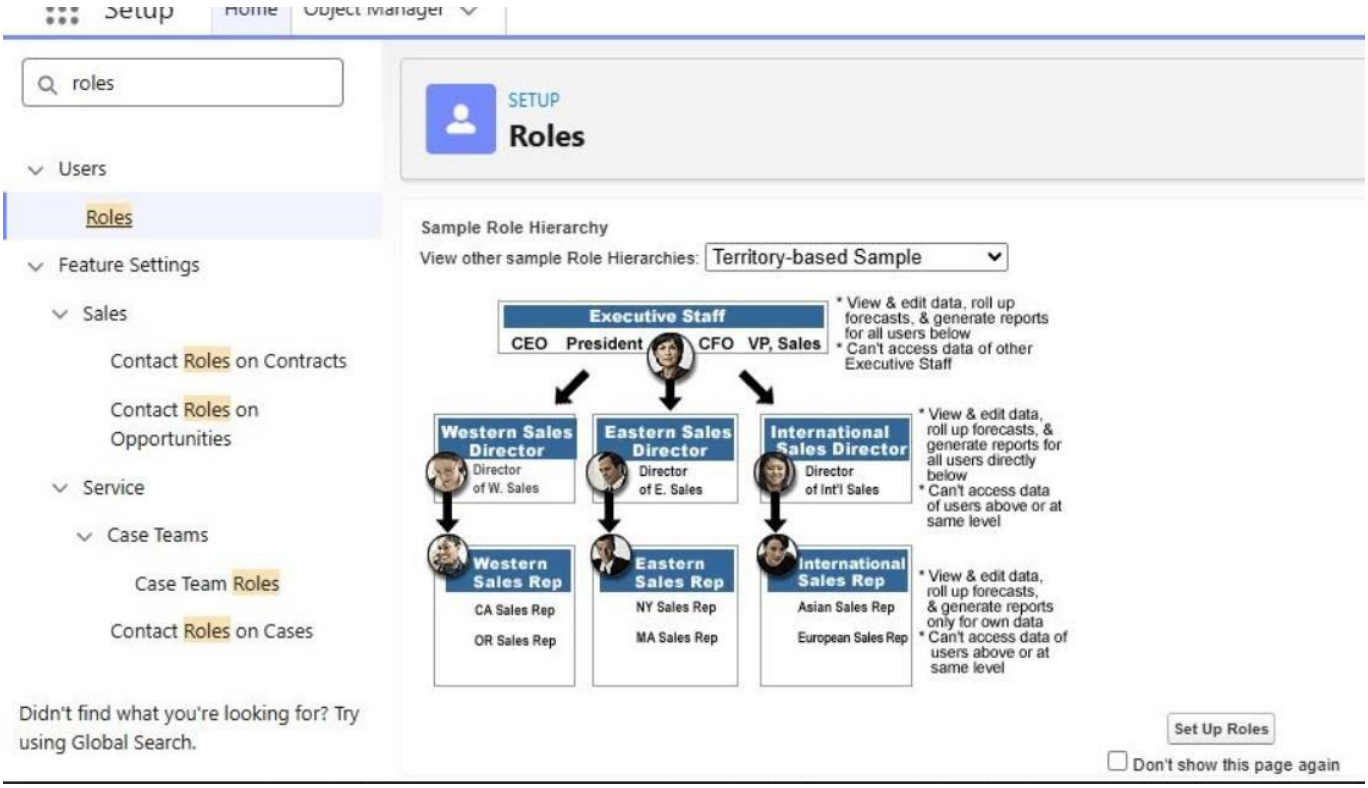
In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity 1: Creation of Courier Admin Role

Creation of Role:

1. From the Quick Find box, search for Roles and click on the Roles option.

2. Select Set Up Roles.



3. Click on Expand All and click Add Role under CEO.
4. Give Label as “Courier Admin”.
    - The Role Name will auto-populate.
5. Click Save.

Activity 2: Creating Branch Manager Role Under Courier Admin Role

Create another role:

1. Create a role with the name “Branch Manager”.
2. Below the Courier Admin, click on Add Role.
3. Enter the label name as “Branch Manager” and the Role Name will automatically populate.
4. Click Save.

Activity 3: Creating Delivery Agent Role Under Branch Manager Role

Steps:

1. Below the Branch Manager role, click on Add Role.
2. Enter Label as “Delivery Agent”.
    - Role Name will auto-populate.
3. Click Save.

Activity 4: Creating Customer Support Role Under Branch Manager

Steps:

1. Under the Branch Manager role, click on Add Role.
2. Enter Label as “Customer Support”.
3. Role Name will automatically populate.
4. Click Save.

Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking

This hierarchy ensures proper data sharing and security in the Courier Management System.

11. Reports & Dashboards

Reports and Dashboards help analyze business data and track courier operations efficiently.

Activity 1: Create Report Types

Step 1: Open Report Types

1. Click Gear Icon → Select Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.

Step 2: Create Shipment Report Type

- Primary Object: Shipment__c
- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 3: Create Customer Report Type

- Primary Object: Customer__c
  - Label: Customer Summary Report
  - Category: Reports
  - Deployment Status: Deployed
  - Click Next → Save

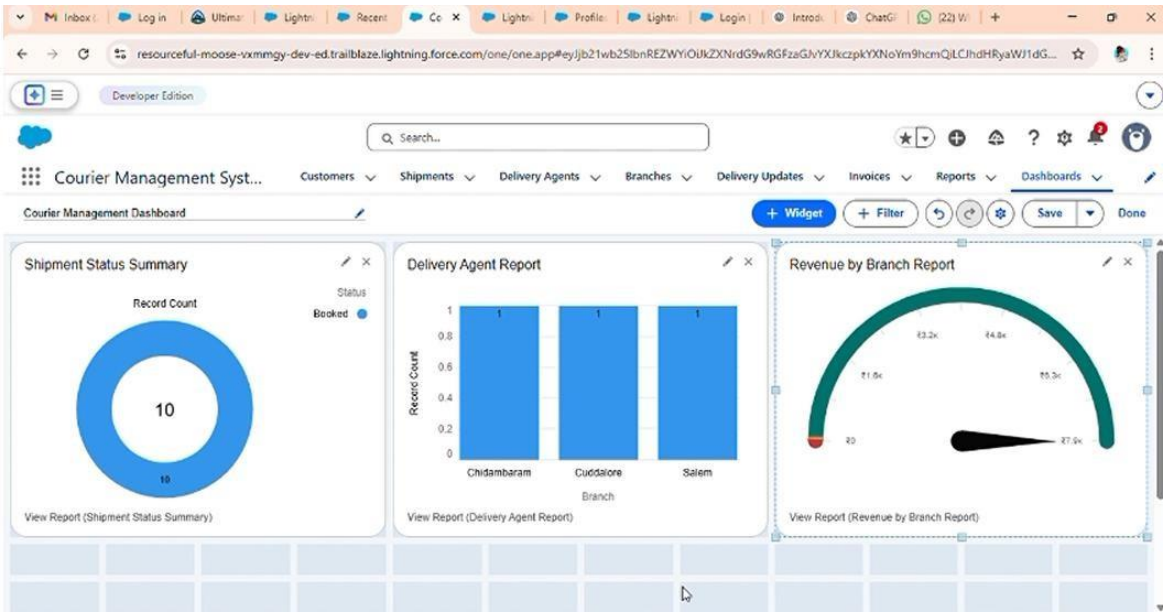
Activity 3: Create Dashboard

Step 1: Open Dashboards

- Click App Launcher (9 dots)
- Search Dashboards
- Click New Dashboard

Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create



Activity 4: Add Dashboard Components

Component 1: Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

Component 2: Agent Performance (Bar Chart)

- Select report: Agent Performance Report
- Chart Type: Bar Chart
- Click Add

Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

Activity 5: Save & Activate Dashboard

1. Click Save
2. Click Done

Share Dashboard With:

- Courier Admin
- Branch Managers
- Customer Support Team

12. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

13. Conclusion:

The Courier Management System successfully digitalizes courier operations by utilizing Salesforce automation and cloud technology. The system simplifies shipment management, improves delivery tracking, and enhances communication with customers. As a result, courier companies can achieve higher efficiency, improved transparency, and better customer satisfaction.

14. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

15. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org